# Lab Experiment 5: JavaScript Objects, Exception Handling, and AI Assistance

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## Learning Outcomes

By completing this lab experiment, students will be able to:

- Create and use JavaScript objects
- Define object properties and methods
- Use the Math.pow() method to perform mathematical calculations
- Create JavaScript functions with parameters and return values
- Apply exception handling using try-catch
- Detect and handle runtime errors in JavaScript
- Use AI tools to assist with coding, debugging, and understanding JavaScript concepts

## Objective

This lab experiment introduces students to JavaScript objects, object methods, mathematical operations using the Math object, exception handling with try-catch, and the use of AI tools to support programming tasks.

Students may use AI tools to:

- Generate code ideas
- Explain JavaScript concepts
- Debug syntax and logic errors
- Improve code readability
- Create additional test cases

### Information Header Example

```
/*
Name: John Smith
Course: JavaScript Programming
Lab Experiment: Objects, Exception Handling, and AI Assistance
Date: May 24, 2026
*/
```

## Class examples

### Class Example 1: JavaScript Object

### Example Code

```
console.log("\n------ Class Example 1: JavaScript Object -----");

const car = {
    brand: "Toyota",
    year: 2025,
    displayInfo: function () {
    return this.brand + " " + this.year;
    }
};

console.log(car.displayInfo());
```

### Class Example 2: Math Object

### Example Code

```
console.log("\n------ Class Example 2: Math Object -----");

let result = Math.pow(4, 2);
console.log("4 squared =", result);
```

### Class Example 3: Exception Handling

### Example Code

```
console.log("\n------ Class Example 3: Exception Handling -----");
try {
    let value = number;
} catch (error) {
    console.log("An error occurred.");
}
```

## Lab Exercises

### Lab Exercise 1: JavaScript Object with AI Assistance

### Description

- Create a JavaScript object named `mycalculator` with two properties and two methods.
- Use AI assistance to help explain the formulas or debug the object methods if needed.

### Object Requirements

The object must contain the following properties:

- **message**, type: String
- **description,** a text message

The object must contain the following methods:

- Method: **`area_square(),`** description: Calculates the area of a square using `Math.pow(side,2)`
- Method: **`volume_cube(),`** description: Calculates the volume of a cube using `Math.pow(side,3)`

**Formula**

- Area of a square = side²
- Volume of a cube = side³

Each method must return the calculated value.

**Example Output**

```
Area of square: 4
Volume of cube: 8
```

## AI Assistance

Open an AI assistant and ask the following questions in the prompt:

- Explain how `Math.pow()` works in JavaScript.
- How do you calculate powers in JavaScript using `Math.pow()`?

**Lab Exercise 2: Exception Handling with AI Assistance**

**Description**

Define a JavaScript function named `readProperty` that takes two parameters:

- obj → an object
- prop → a property name

The function should attempt to access the property from the object. If an error occurs (for example, when accessing a property from null or undefined), the function must catch the error and return the message:

"Error accessing property"

The program must use `try-catch` inside the function.

Use AI assistance to help debug the `try-catch` statement in Lab Exercise 2, the sample code below, and explain how exception handling works.

**Example Output**

```
John
Error accessing property
```

**Sample Code**

```
console.log("\n------ Lab Exercise 2: Exception Handling -----");

function readProperty(obj, prop) {
    try {
        return obj[prop];
    } catch (error) {
        return "Error accessing property";
    }
}

// Example 1
const student = {
    name: "John",
    age: 20
};

console.log(readProperty(student, "name"));

// Example 2
console.log(readProperty(null, "name"));
```

// AI Assistance:
// ChatGPT helped debug the try-catch
// statement and explain runtime errors.

**AI Reflection Activity**

**Instructions**

After completing the lab, add a short comment using /* */ at the end of the .js file describing how AI assisted you during the activity.

Example:

/* Reflection:
1) AI helped me understand object methods
2) exception handling, and debugging errors.
*/

**Submission Requirements**

- Complete the class examples and lab exercises in one .js file
- Upload the completed .js file to your GitHub repository
- Submit the GitHub repository HTTPS link

**Grading Criteria**

- Class examples: 2 pts
- Lab exercise 1: 4 pts
- AI assistance: 4 pts